Experiment:-5

Problem statement:-

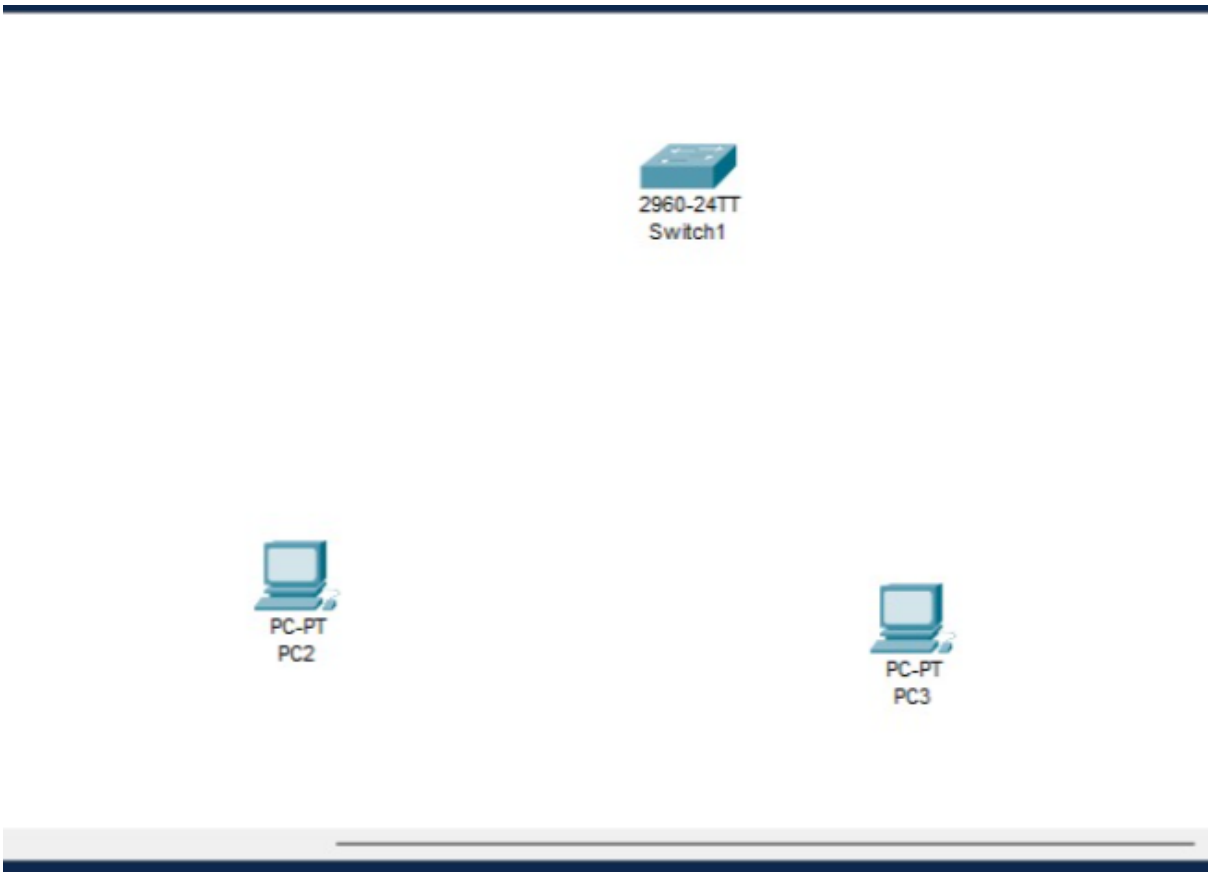
Design and simulate flow control and error control protocols such as Stop and Wait, Go-Back-N ARQ, and Selective Repeat ARQ using Cisco Packet Tracer. Compare their performance in terms of throughput and efficiency under varying network conditions.

Aim:- Simulation of Flow Control and Error Control Protocols

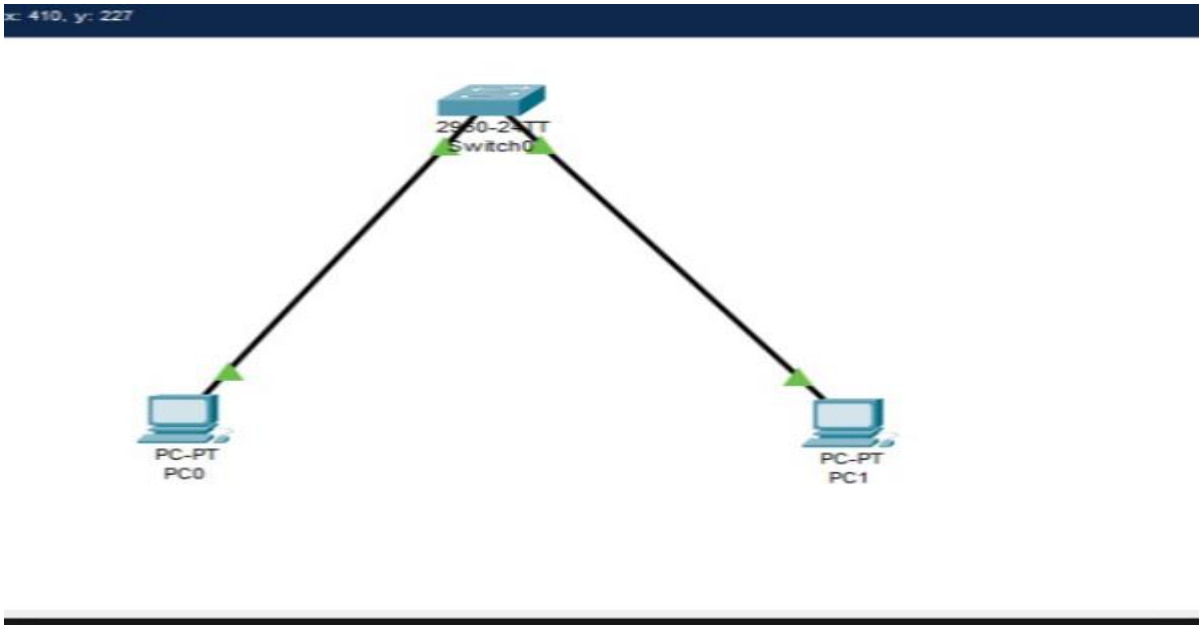
Procedure:-

Step 1: Setup the Network Environment

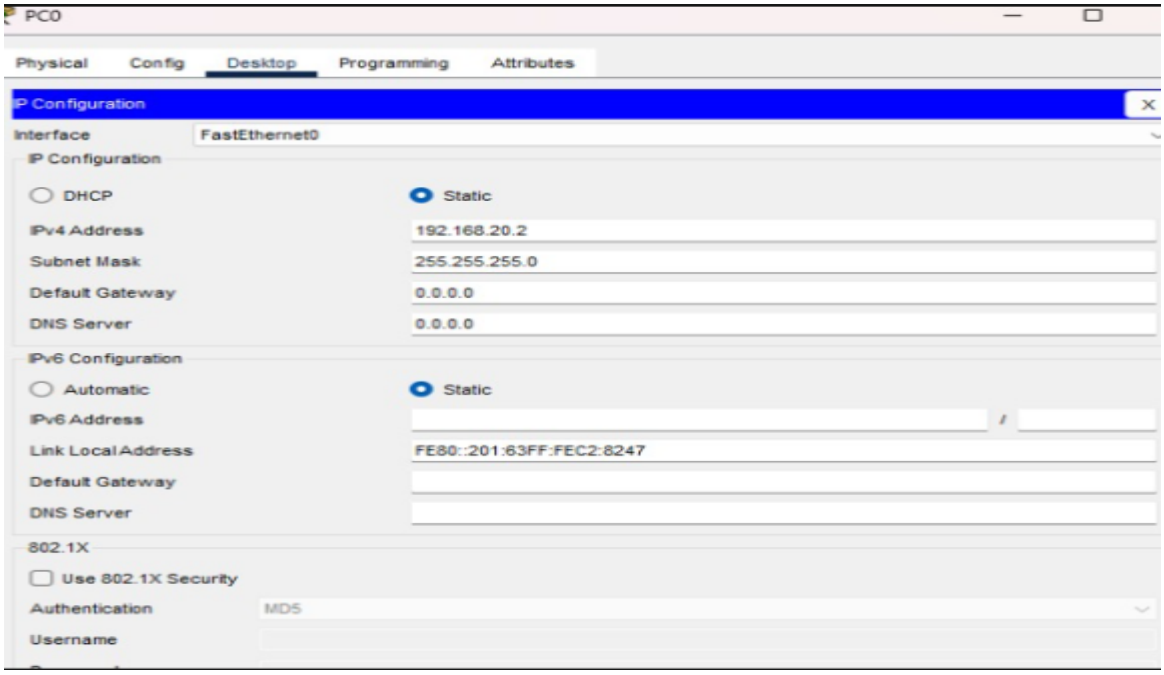
- Open **Cisco Packet Tracer**
- Click **File → New**
- Add devices:
  - 2 PCs (PC0 – Sender, PC1 – Receiver)
  - 1 Switch (2960)



- Connect:
  - PC0 → Switch
  - PC1 → Switch
- Use **Copper Straight-Through Cable**



Step 2: Configure IP Addresses



Step 3: Test Connectivity

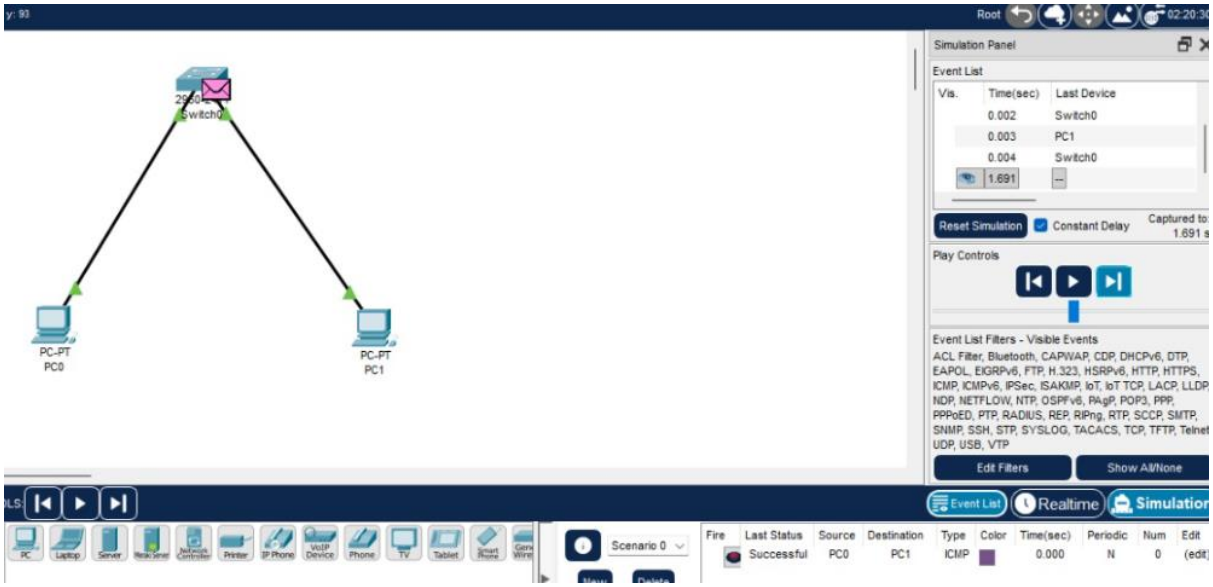
- Open Command Prompt on PC0
- Type:

```
ping 192.168.1.2
```

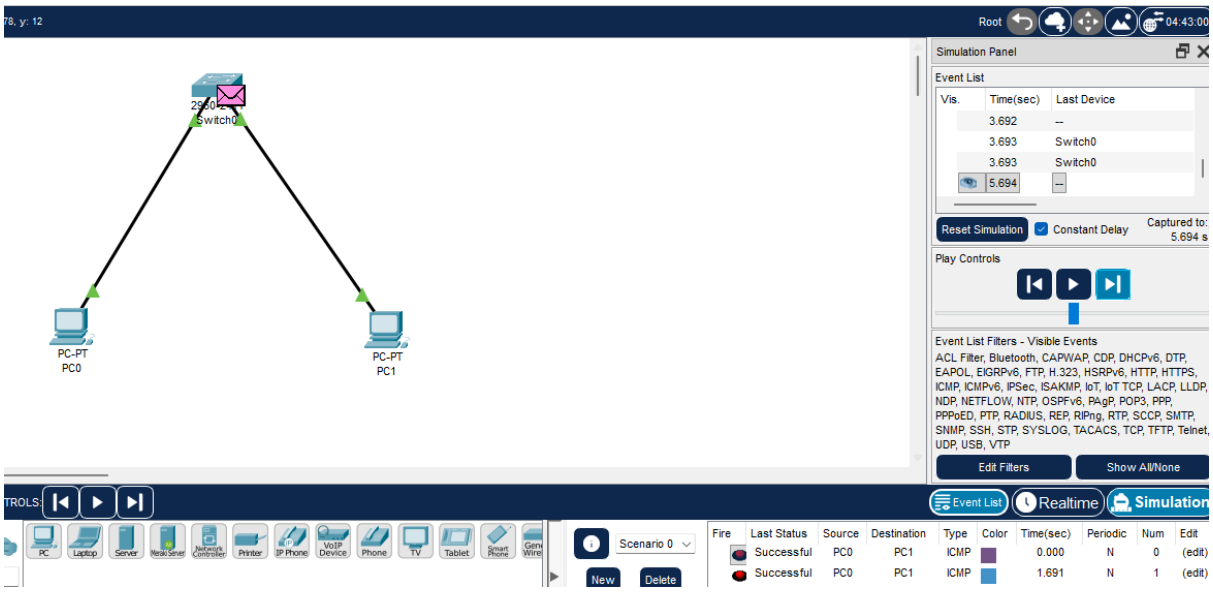
- Ensure successful replies

```
C:\>ping 192.168.20.3

Pinging 192.168.20.3 with 32 bytes of data:

Reply from 192.168.20.3: bytes=32 time<1ms TTL=128
Reply from 192.168.20.3: bytes=32 time<1ms TTL=128
Reply from 192.168.20.3: bytes=32 time<1ms TTL=128
Reply from 192.168.20.3: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.20.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

Step 4: Simulate Stop and Wait ARQ



Step 5: Simulate Go-Back-N ARQ



Stop and Wait ARQ

- Sender transmits **one frame at a time**
- Waits for **ACK (acknowledgment)** before sending next frame
- If ACK is not received → **frame is retransmitted**
- Uses **simple mechanism**
- Low efficiency due to waiting time
- Suitable for **small data transmission**
- No pipelining (only one frame in transit)

Go-Back-N ARQ

- Sender can send **multiple frames continuously** (window size > 1)
- Uses **sliding window protocol**
- Receiver accepts only **in-order frames**
- If one frame is lost/error:
    - Sender retransmits **that frame and all subsequent frames**
- Improves efficiency compared to Stop & Wait
- Wastes bandwidth due to multiple retransmissions
- Used in reliable data transfer systems

Selective Repeat ARQ

- Sender sends **multiple frames (like Go-Back-N)**
- Receiver accepts **out-of-order frames**
- Only the **erroneous/lost frame is retransmitted**
- Uses **separate acknowledgment for each frame**
- More efficient than Go-Back-N
- Requires more memory and complexity
- Minimizes unnecessary retransmissions